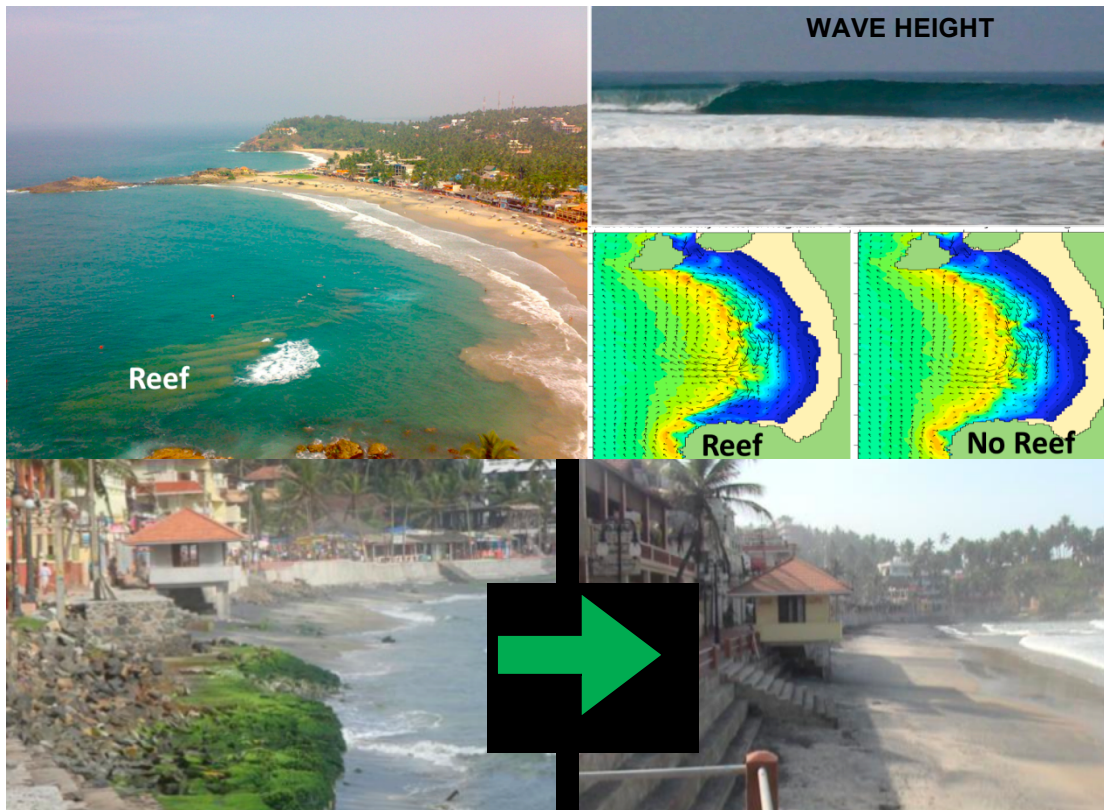

Location	Kovalam, Kerala, India
Type of Project	Multi-Purpose Reef, Beach Protection
Client	Harbour Engineering Department, Government of Kerala, India



Clockwise from upper left: aerial view of the Kovalam Multi-Purpose Reef; high quality surfing wave breaking along the reef; wave height models after and before construction illustrate how the reef both shapes surf and redirects erosive wave energy offshore.

Project Overview – Kovalam has been one of Southern India’s most popular tourist destinations since the 1930’s. However due to rapid development in recent years, the Kovalam shoreline has suffered from severe coastal erosion during the monsoon season which has led to the loss of a valuable beach amenity. The Kerala Department of Tourism considered traditional management options including seawall construction and managed retreat, but ultimately elected to apply a more sustainable solution to this serious problem. ASR was contracted to design and construct an environmentally sensitive coastal protection structure at Kovalam.

Project Outcomes – After conducting extensive field studies, developing numerical and physical models, and performing exhaustive calibrations and testing, ASR completed construction of the Kovalam Multi-Purpose Reef in February 2010. The reef has been specifically designed to cause the region’s powerful waves to break offshore, thereby minimizing the erosive effects of those

waves on the beach. Additionally, the reef is positioned to redirect the currents that have been partially responsible for the historical reduction of beach size in Kovalam.

Local monitoring reports show that the reef has been hugely successful in creating a wider, more stable section of beach in Kovalam. The reef is a showcase in the concept of sustainable managed advance as the valuable tourist beach and structures behind it have successfully been protected without damaging the local environment. The reef has also stimulated tourism in the region enhancing existing surfing waves, and has been featured by organizations including Patagonia, Surfline, Quicksilver, and others. ASR and the Kerala Department of Tourism continue to monitor reef performance, and ASR is working closely with the local stakeholders to maximize the reef's long-term benefit to the community.